CSCE 411: Design and Analysis of Algorithms (Honors)

Week 11: Reductions and Undecidability

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1 Vertex Cover Problem

A vertex cover of an undirected graph $G = (V, E)$ is a set of nodes $S \subseteq V$ such that:

We say an edge is _____ if at least one of its vertices are in the vertex cover.

The **vertex-cover problem** is the task of finding a vertex cover of minimum size. The decision version of the problem is written as follows:

VertexCover =

Theorem 1.1. *The vertex-cover problem is* _____

Question 1. *Assume we have proven the above theorem. Which of the following statements is true?*

A *Vertex-cover is NP-complete, because any problem that is NP-hard is also NP-complete*

B *Vertex-cover is NP-complete, because of a different reason from the above reason*

C *It is possible that vertex-cover is not NP-complete, this is still an unsettled open question*

D *The fact that vertex-cover is NP-hard means that we can for sure rule out the possibility of having a polynomial-time algorithm for solving it.*

The complement of a graph

Given an undirected graph $G = (V, E)$, the *complement* graph $\bar{G} = (V, \bar{E})$ is the graph with the same set of nodes where:

Formally, this means that:

$$\bar{E} =$$

Observation. The complement of $\bar{G}$ is G. In other words, taking the complement of a complement returns the original graph.

Proof that vertex-cover is NP-hard

We will reduce the Clique Problem to vertex cover.

The reduction

- Let $G = (V, E)$ be the input for $\textsc{Clique}(G, k)$
- Construct the complement graph $\bar{G}$
- The output is $\langle \bar{G}, |V| - k \rangle$; i.e.,

What must hold for the reduction

- ($\implies$) Assume G has a k-clique and prove this means $\bar{G}$ has a vertex cover of size $|V| - k$
- ($\impliedby$) Assume $\bar{G}$ has a vertex cover of size $|V| - k$ and prove that G has a k-clique

We could prove each piece separately, but in some cases there is an easier way to show a reduction all at once.

Theorem 1.2. *A graph G has a size k clique $\iff$ $\bar{G}$ has a size $|V| - k$ vertex cover.*

2 Other NP-complete problems

We have already talked about the following NP-complete problems

- SAT and 3SAT
- The CLIQUE problem
- The VERTEX COVER problem

Although we will not provide explicit reductions to prove the following are NP-complete, here are some more well-studied NP-complete problems you should be aware of:

- Subset sum
- Maximum Cut
- Hamitonian Path
- Traveling Salesman
- Graph Coloring
- Clique Cover

Warm-up question before we move on

Question 2. *It is NP-hard to find a vertex cover for a graph G*

A *True*

B *False*

3 Summary of a few NP-complete graph problems

Subset sum The input to a subset sum problem is a list of integers L (could be positive or negative) and a target sum T. The goal is to find a subset of integers in L that sum to T. Several special variants are also NP-complete:

- The special case where $T = 0$
- The special case where all integers are positive
- The special case where all integers are positive and $T = \frac{1}{2}\text{sum}(L)$.

All graph problem in the rest of the lecture take as input an undirected graph $G = (V, E)$ (which may or may not have weights) and an arbitrary nonnegative integer k.

Hamiltonian Cycle A Hamiltonian cycle is a cycle in the graph that visits every node exactly once.

Traveling salesman problem The traveling salesman problem seeks the smallest weight Hamiltonian cycle. The decision version asks: is there a Hamiltonian cycle with weight at most W?

Graph Coloring Check whether we can assign one of k colors to each node $v \in V$ in such a way that if $(u, v) \in E$, then u and v have been assigned different colors.

This problem is NP-hard even if $k = 3$, but becomes polynomial time for $k = 2$.

Clique Cover A clique cover of size k is set of nodes $\mathcal{S} = \{S_1, S_2, \ldots S_k\}$ such that:

- $\mathcal{S}$ is a partition: $S_i \cap S_j = \emptyset$ if $i \neq j$, and $\bigcup_{i=1}^{k} S_i = V$
- S_i is a clique for each $i = 1, 2, \ldots, k$

Checking whether a graph G has a size k clique cover is NP-complete

4 Relationship between clique cover and graph coloring

Recall the following theorem regarding the relationship between the Vertex Cover problem and the Clique problem.

Theorem 4.1. *A graph $G = (V, E)$ has a clique of size k if and only if the complement graph $\bar{G}$ has a vertex cover of size $|V| - k$.*

Proof.

There is a relationship between graph coloring and clique cover that is very similar and can be proven using similar arguments. This will be a good practice problem for the final exam!

Maximum cut A *cut set* of an undirected graph G is a partition of the nodes $\{S, V - S\}$ and the cut value is the weight of edges across the cutset:

$$\mathbf{cut}(S) = \sum_{(u,v)\in E} w_{uv}. \tag{1}$$

Finding a set S that maximizes $\mathbf{cut}(S)$ is the *maximum cut* problem.

The decision version asks: given a value k, is there a set S with $\mathbf{cut}(S) \geq k$?

5 Decision problems and undecidability

A decision problem Q is *undecidable* if there cannot exist an algorithm that always provides the correct YES or NO answer.

Question 3. *Say that Q is an NP-Complete problem. Which of the following statements is true?*

A *Q is undecidable*

B *If $P \neq NP$, then Q is undecidable, but we just do not now for sure yet*

C *Neither of the above is true*

6 Halting and the Halting Problem

Let P represent some notion of a computer program and I represent some type of input that you give P. You can think of both as strings.

We say that P *halts* on I if _____

Example Let $P = \text{OddHello}$ be represented by the following code snippet.

```
OddHello(I)
  if the input is a string with an even number of letter characters (a-z) then
      while true do
          continue
      end while
  else
      return "hello!"
  end if
```

Question 4. *Does $\langle P, I \rangle$ halt when $I =$ "hello"?*

A *Yes*

B *Nope*

C *It depends*

D *There is no way of knowing*

Question 5. *Does $\langle P, I \rangle$ halt when $I =$ "hi"?*

A *Yes*

B *Nope*

C *It depends*

D *There is no way of knowing*

Question 6. *Does $\langle P, I \rangle$ halt when $I = P$?*

A *Yes*

B *Nope*

C *This question cannot have an answer*

Definition: The Halting Problem. The *halting problem* is a decision problem that takes in inputs of the form $\langle P, I \rangle$ and outputs

- YES

- NO

To be clear

- An instance of the halting problem is given by a

- An algorithm A solves the halting problem if

Theorem 6.1. *The Halting Problem is* _____